FINANCIAL ANALYSIS PORTFOLIO

Financial modeling and cash-flow forecasting

Northstar Industrial Components

A traceable work sample: monthly revenue analysis, linked financial statements, scenario testing and discounted cash-flow valuation.

Yazhu Shen

96 monthly observations | 2026–2030 projections

Fictional company and simulated data. Prepared as a skills demonstration.

From monthly data to a valuation

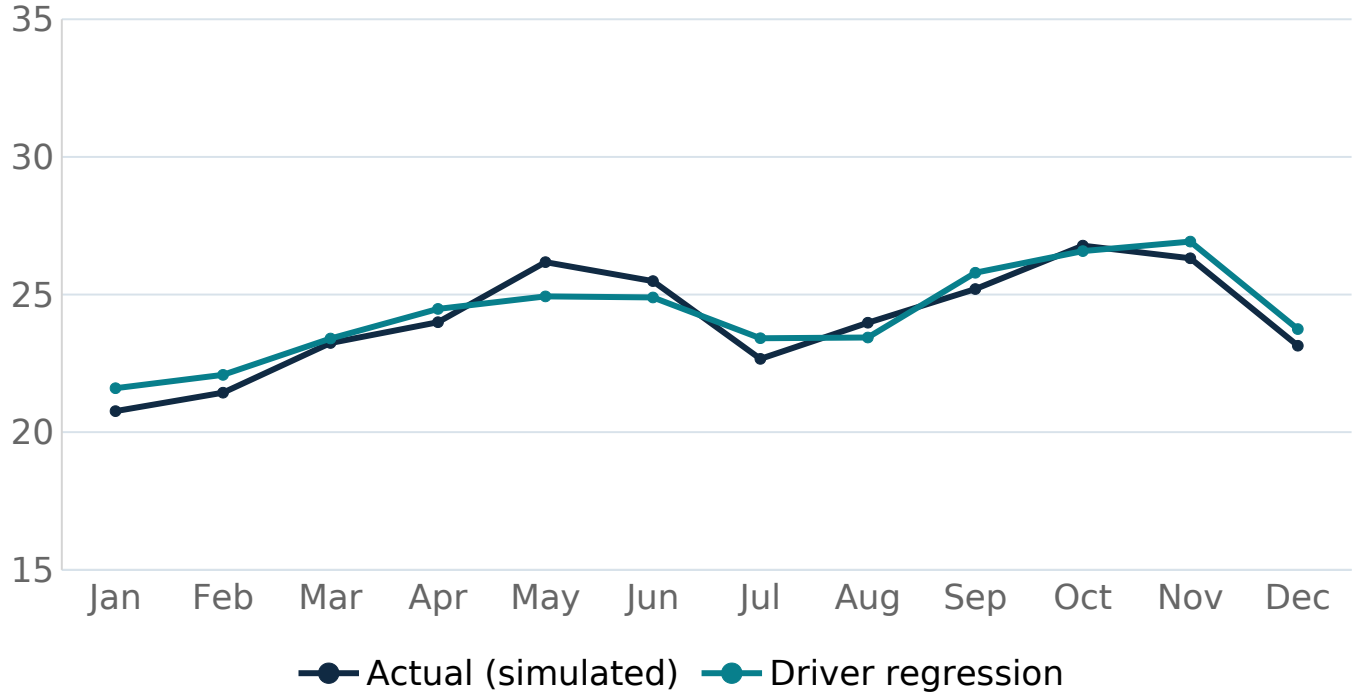
Each stage has a visible input, output and review point.

Stage	Model output	Review point
1 Prepare the data	96 monthly observations 2018–2025	Dates, units and duplicate checks
2 Compare forecasts	Driver regression, trend/seasonal and seasonal-naive benchmark	84 training months; 12 holdout months
3 Build the financial model	Linked income statement, balance sheet and cash-flow projections	Cash roll-forward and balance checks
4 Value and stress-test	Unlevered cash flows, DCF and Base/Downside scenarios	WACC, terminal growth and liquidity

2026 revenue uses the selected statistical model. 2027–2030 growth rates are explicit scenario assumptions.

Driver regression has the lowest holdout error

2025 conditional validation • monthly revenue, USD millions



Model	RMSE
Driver regression	0.662
Trend + seasonality	1.034
Seasonal naive	2.149

Driver MAPE

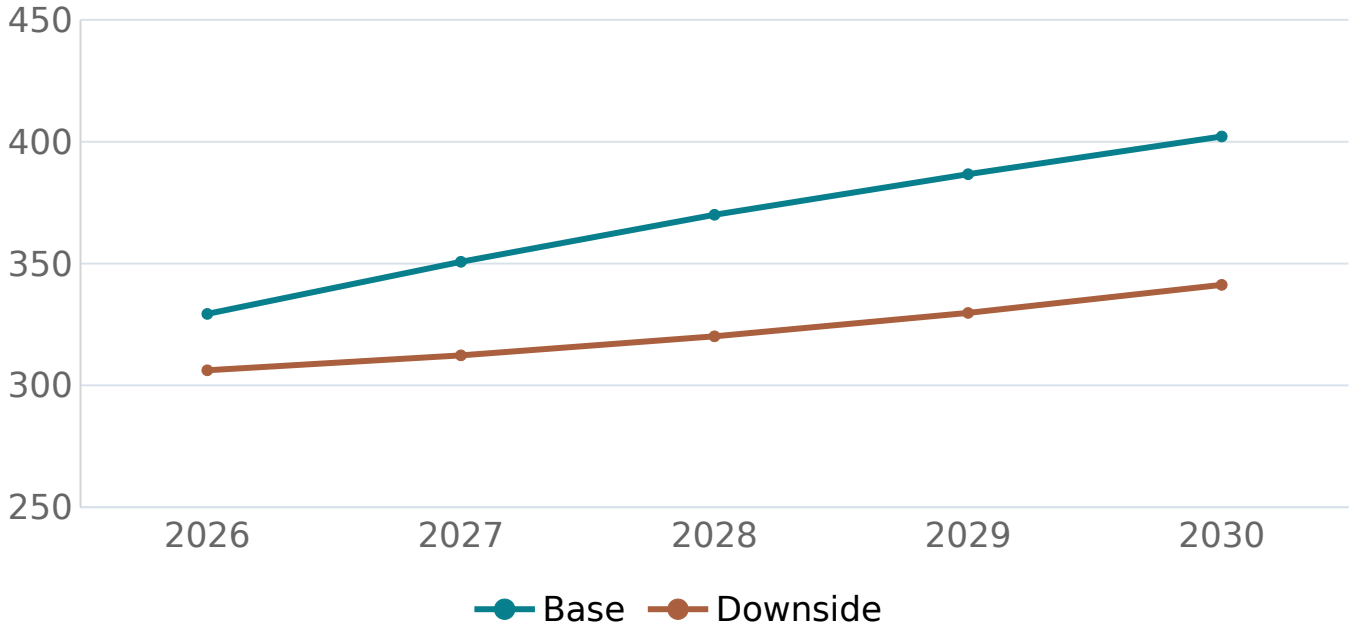
2.53%

The driver model uses realized holdout index values. This measures conditional accuracy; additional rolling-window tests are needed before operational use.

Downside enterprise value falls 31.1%

Scenario forecasts • USD millions • valuation date 31 December 2025

Revenue projection



DCF output	Base	Downside
Enterprise value	563.2	388.0
Equity value	540.5	365.2
2026 free cash flow	27.4	3.4
Terminal share of EV	74.3%	78.2%

Both cases: WACC 9.5%; terminal growth 2.5%.

Slower growth, lower margins and weaker working-capital efficiency drive the downside.
Terminal value contributes 74.3% of base EV, making long-run assumptions especially important.

A model reviewers can inspect and rerun

Editable deliverables and a reproducible reference analysis

Tool / artifact	Purpose	Execution status
Excel workbook	Linked statements, DCF, scenarios	Formula checks supplied with model
Python reference analysis	Regression, holdout tests, model checks	Executed; outputs included
Stata / EViews scripts	Equivalent estimation and forecast steps	Supplied; vendor runtimes not run
Power Query M script	Import and prepare source data	Supplied; native refresh not run
Word / PowerPoint	Explain methods and decision implications	Editable documents included

Suggested review

Switch the workbook scenario. Inspect the cash and balance checks. Trace the DCF to free cash flow. Rerun the Python analysis, then compare the exported metrics with the presentation.